

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO1: Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1,BL2
Assessment Tool Weightage: 30%

QUESTIONS: 2

Total Marks:50

Annexure A

Technical Presentation

<i>Criteria</i>	<i>Technical Content Accuracy(3)</i>	<i>Problem Identification(2.5)</i>	<i>Use of Tools(2)</i>	<i>Communication(1.5)</i>	<i>Professionalism(1)</i>	<i>Total(10)</i>
<i>Weightage</i>	30%	25%	20%	20%	10%	100%
<i>Q1</i>						
<i>Q2</i>						

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Technical Presentation					
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Annexure A

Topic 1: Decision Support Systems (DSS)

Focus Areas:

- Definition and need of DSS
 - Components (Data, Model, UI)
 - Types of DSS
 - Real-world applications (banking, healthcare)
 - Advantages & limitations
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Topic 2: Operational Systems vs DSS

Focus Areas:

1. OLTP vs OLAP concepts
 2. Key differences (data, users, purpose)
 3. Performance comparison
 4. Use-case scenarios
 5. Integration between systems
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Topic 3: Architecture of Data Warehouse

Focus Areas:

1. Data warehouse definition
 2. Three-tier architecture
 3. ETL process
 4. Data marts & metadata
 5. Benefits in decision-making
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Topic 4: OLAP and Data Cubes

Focus Areas:

1. OLAP concept
 2. Data cube structure
 3. OLAP operations (slice, dice, roll-up, drill-down)
 4. Applications in analytics
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Topic 5: Dimensional Data Modeling

Focus Areas:

1. Fact and dimension tables
 2. Star schema vs snowflake schema
 3. Schema design process
 4. Real-world example
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Topic 6: Query Processing & BI Tools

Focus Areas:

1. Query processing stages
 2. Query optimization
 3. BI tools overview
 4. Reporting and visualization
 5. Examples of open-source BI tools
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Topic 7: OLAP Application using KNIME

Focus Areas:

1. Introduction to KNIME
2. Workflow components (nodes, connections)
3. Performing OLAP operations
4. Sample application/demo

5. Report generation
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PART B: PRESENTATION RULES & GUIDELINES

◇ 1. Group Formation

1. Each group: 3–5 students
 2. Each member must participate
 3. Assign roles:
 1. Presenter
 2. Designer (PPT)
 3. Researcher
 4. Demo/Case handler
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◇ 2. Presentation Structure (MANDATORY)

Each group must include:

1. Introduction
 1. Definition and importance
 2. Core Concepts
 1. Explanation with diagrams
 3. Process/Architecture
 1. Step-by-step explanation
 4. Real-World Example
 1. Case study or application
 5. Tool/Demo (if applicable)
 1. KNIME or BI tool
 6. Conclusion
 1. Summary + future scope
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◇ 3. Slide Preparation Rules

1. Maximum slides: 10–15
2. Use:
 1. Diagrams
 2. Flowcharts

3. Tables (for comparison)
 3. Avoid:
 1. Too much text
 2. Copy-paste content
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◇ 4. Presentation Delivery Rules

1. Time limit: 10–12 minutes per group
2. Each member must speak at least 2 minutes
3. Maintain:
 1. Eye contact
 2. Clear voice
 3. Logical flow



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Unlocking Business Intelligence with OLAP

CSA1607-Data warehouse and Data mining

Supervisor:
Dr. Beulah David

Presented By:
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Introduction to OLAP

- OLAP stands for **Online Analytical Processing**.
- It helps analyze large volumes of business data quickly.
- Supports decision-making by providing multidimensional data analysis.
- Used by businesses to identify trends, patterns, and performance.
- Commonly used in retail, banking, healthcare, and e-commerce.



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Understanding the Data Cube

- A Data Cube organizes data into multiple dimensions.
- Common dimensions include **Time, Product, and Location**.
- Enables users to view data from different perspectives.
- Improves speed and efficiency of data analysis.
- Helps compare sales, profits, and business performance easily.



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OLAP Operations

- Four Powerful OLAP Operations
- **Slice:** Displays data for one specific dimension.
- **Dice:** Filters data using multiple dimensions.
- **Drill Down:** Moves from summary data to detailed data.
- **Roll Up:** Summarizes detailed data into higher-level information.
- These operations enable flexible and interactive business analysis.



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Real-World Applications of OLAP

- Retail companies analyze customer purchasing behavior.
- Banks detect fraud and monitor financial transactions.
- Hospitals analyze patient records and treatment outcomes.
- Educational institutions evaluate student performance.
- E-commerce companies study customer preferences and sales trends.



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Benefits of OLAP

- Provides fast and accurate data analysis.
- Improves strategic and operational decision-making.
- Identifies trends and business opportunities.
- Generates interactive reports and dashboards.
- Saves time by processing large datasets efficiently.

THANK YOU